CSC1058 USER INTERFACE DESIGN & IMPLEMENTATION

MODULE COORDINATOR & LECTURER
Hyowon Lee

TEACHING ASSISTANT
Haoyan Zhang

	Tuesday (3 – 4pm)	Wednesday (11am - 12pm)
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Think about why this title slide is inferior to the previous slide, in terms of information/administration ratio

USER-INTERFACE DESIGN AND IMPLEMENTATION

WEEK 2 – DESIGN GUIDELINES

2

"Good" UI

A: "Wow, this website is stunning."
B: "Well, not to me…"
A: "This app is very easy to use."
B: "Really? I couldn't figure out how to…"
"That's my favourite gadget, because…"

"I think it needs to be more user-friendly"

"This online booking is really frustrating ☹"

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"Good" UI

A: "Wow, this website is stunning."
B: "Well, not to me…"
A: "This app is very easy to
B: "Really? I couldn'
"That's m
dget, because…"
ds to be more user-friendly"
This online booking is really frustrating ☹"

Context Diversity

A good UI for me might not be a good UI for you?

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Personality trait influences user's preference, reaction, behaviour and performance

Myers-Briggs Type Indicator (based on Jung Typology)

- Extraversion vs. Introversion
- Sensing vs. Intuition
- Thinking vs. Feeling
- Judging vs. Perceiving

http://www.humanmetrics.com/

Context Diversity

- User difference
  - Learning style
  - **Personality traits**
  - Motor skill
  - Memory
  - Preference
  - Past experience
  - Culture
  - Language
  - ⋮

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Personality trait influences user's preference, reaction, behaviour and performance

Five-Factor Model of personality

- Neuroticism – Sensitive/Nervous vs. Secure/Confident
- Extraversion – Outgoing/Energetic vs. Shy/Withdrawn
- Openness to experience – Inventive/Curious vs. Cautious/Conservative
- Agreeableness – Friendly/Compassionate vs. Competitive/Outspoken
- Conscientiousness – Efficient/Organised vs. Easy-going/Careless

Digman, J. M. (1990). Personality structure: Emergence of the five-factor model. Annual Review of Psychology, 41, 417-440.

Context Diversity

- User difference
  - Learning style
  - **Personality traits**
  - Motor skill
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  - Culture
  - Language
  - ⋮

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Personality trait influences user's preference, reaction, behaviour and performance

E.g.: People with high emotional instability tend to give up information searching more easily*

E.g.: Extravert people tend to have less in-depth analysis on information*

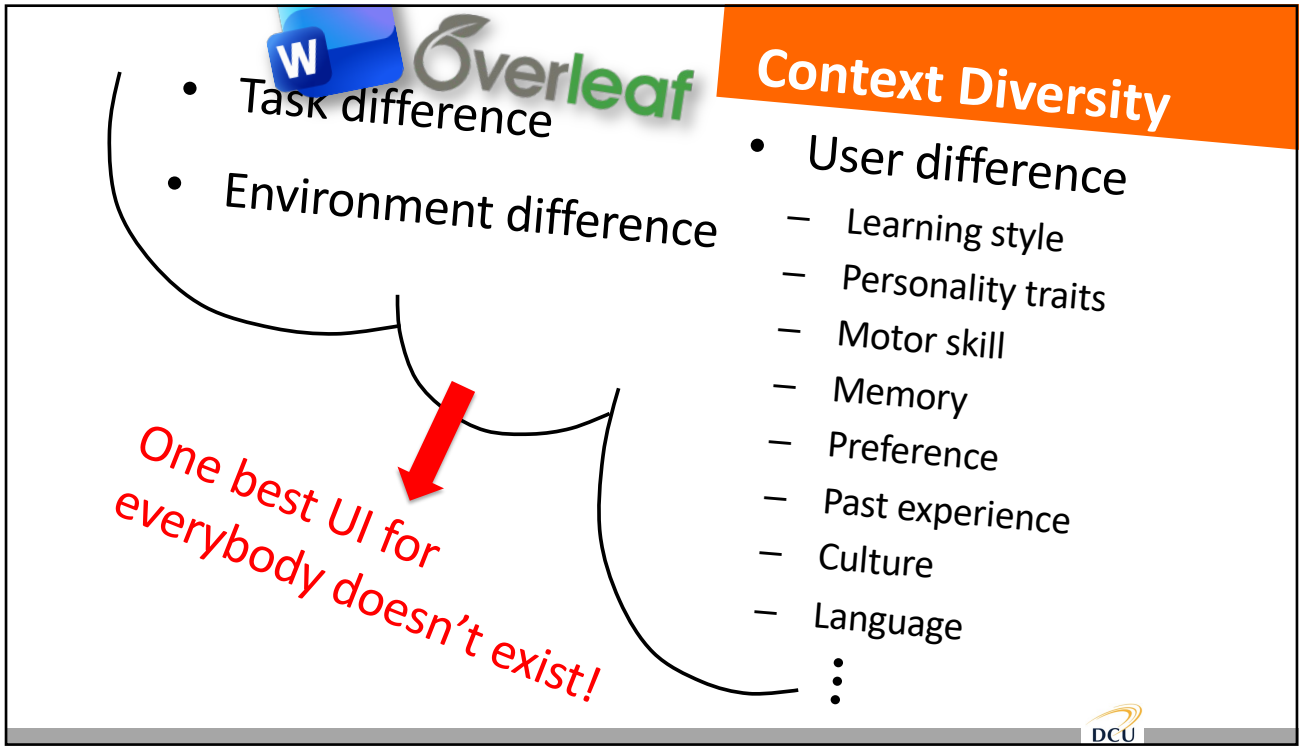
*Long experience in searching reduces the influence of personality

Context Diversity

- User difference
  - Learning style
  - **Personality traits**
  - Motor skill
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  - Past experience
  - Culture
  - Language
  - ⋮

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Context Diversity

- User difference
  - Learning style
  - Personality traits
  - Motor skill
  - Memory
  - Preference
  - Past experience
  - Culture
  - Language
  - ⋮

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"Good" UI

What is a "good" user interface?

➡ "Good" UI is a UI suitable for its intended **target users** trying to do a **specific task** in a **specific environment**

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"Good" UI

Target users

Narrow down to specific group

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User Classification

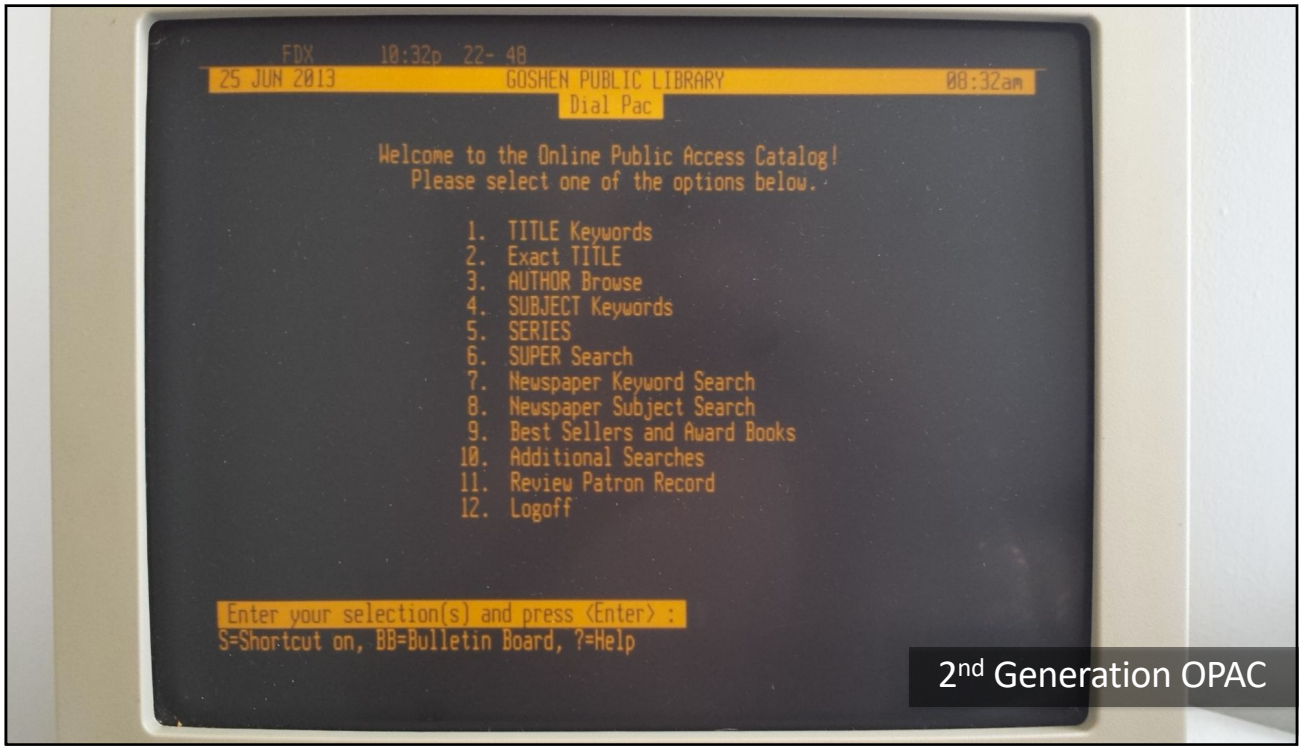
- E.g. a user is writing a report using a word processor…
- E.g. a user is searching for information on medical problem using Google…

Expert user vs. Novice user

… but Expert or Novice in what?

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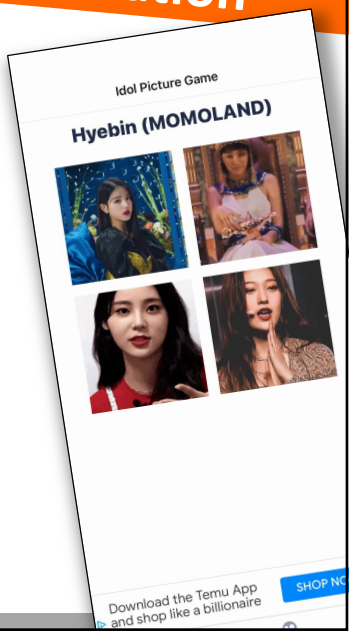
User Classification

… but Expert or Novice in what?

in the familiarity with the app?

in the use of my phone?

in my knowledge on K-pop?



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User Classification

- **Computer skill & general OS knowledge**
  - Window swapping, moving, scrolling, click vs. double-click, swiping, back home, short-cut keys, etc.
- **Knowledge on the software/app in concern**
  - Menu structure, where to find what, within-page searching, etc… i.e. how to use the app
- **Knowledge on the task (= domain knowledge)**
  - Terminology, major topics, authoritative figures and names, etc.

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User Classification

Concider why this table is useful for designers

Computer (OS) skill	Knowledge on the app	Domain knowledge	
GOOD	GOOD	GOOD	→ "Expert user"
GOOD	GOOD	POOR	
GOOD	POOR	GOOD	
GOOD	POOR	POOR	→ "Computer literate"
POOR	GOOD	GOOD	
POOR	GOOD	POOR	
POOR	POOR	GOOD	
POOR	POOR	POOR	→ "Novice user"

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User Classification

Who are your target users?

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General Guidelines

Common sense, research findings, designer's experience, other disciplines… incorporating ***design principles***

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From Jane Eyre, Chapter 1, Charlotte Bronte

There was no possibility of taking a walk that day. We had been wandering, indeed, in the leafless shrubbery an hour in the morning; but since dinner (Mrs. Reed, when there was no company, dined early) the cold winter wind had brought with it clouds so sombre, and a rain so penetrating, that further out-door exercise was now out of the question.

I was glad of it: I never liked long walks, especially on chilly afternoons: dreadful to me was the coming home in the raw twilight, with nipped fingers and toes, and a heart saddened by the chidings of Bessie, the nurse, and humbled by the consciousness of my physical inferiority to Eliza, John, and Georgiana Reed. The said Eliza, John, and Georgiana were now clustered round theirmama in the drawing-room: she lay reclined on a sofa by the fireside, and with her darlings about her (for the time neither quarrelling nor crying) looked perfectly happy. Me, she had dispensed from joining the group; saying, "She regretted to be underthe necessity of keeping me at a distance; but that

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General Guidelines

Common sense, research findings, designer's experience, other disciplines... incorporating ***design principles***

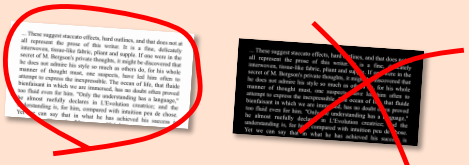
Empirical study into heuristics

User study:

Snyder, Harry L., Decker, Jennie J., Lloyd, Charles J.C., and Dye, Craig. Effect of image polarity on VDT task performance. Proceedings of Human Factors Society - 34th Annual Meeting, Santa Monica, CA, 1990.

Result shows that:

Test readers read faster dark text on light background orientation.



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General Guidelines

Common sense, research findings, designer's experience, other disciplines... incorporating ***design principles***

Empirical study into heuristics

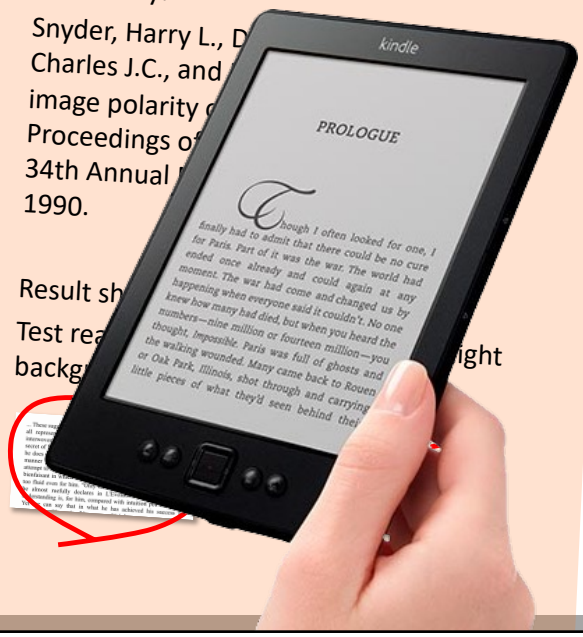
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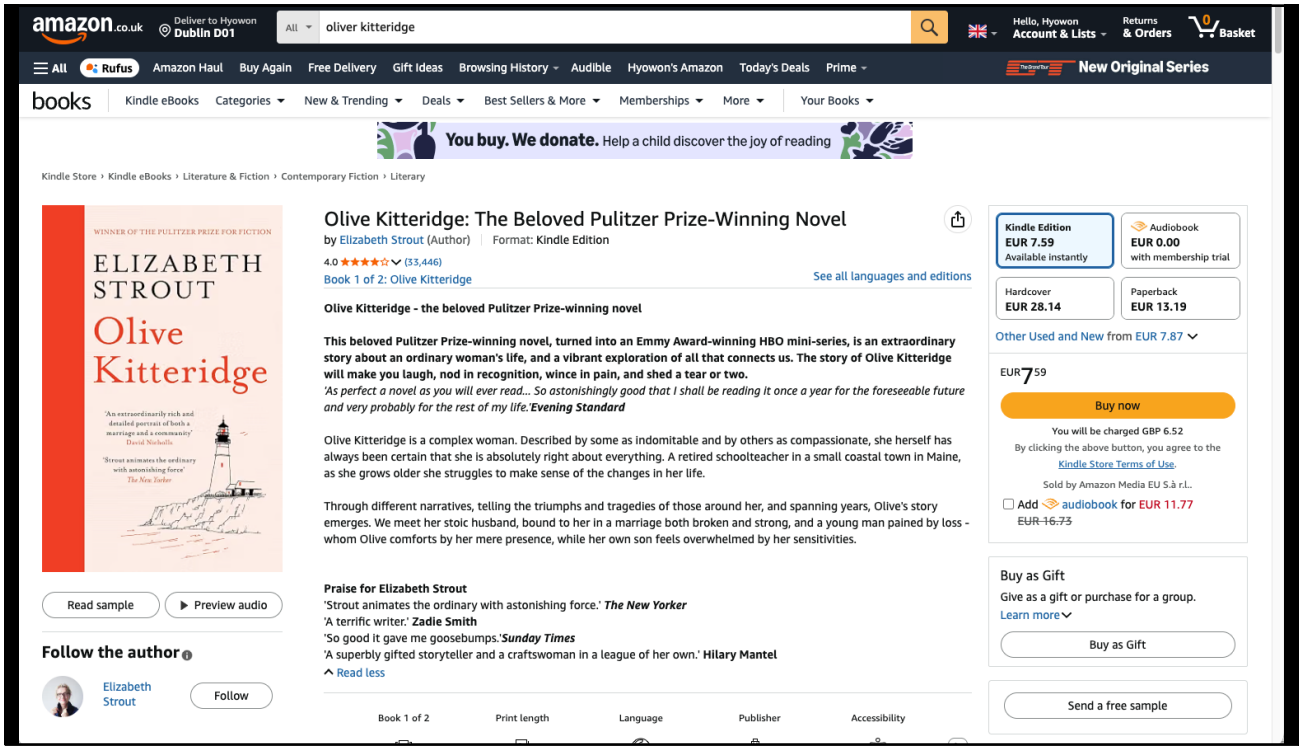
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Designers can simply follow what empirical study results show

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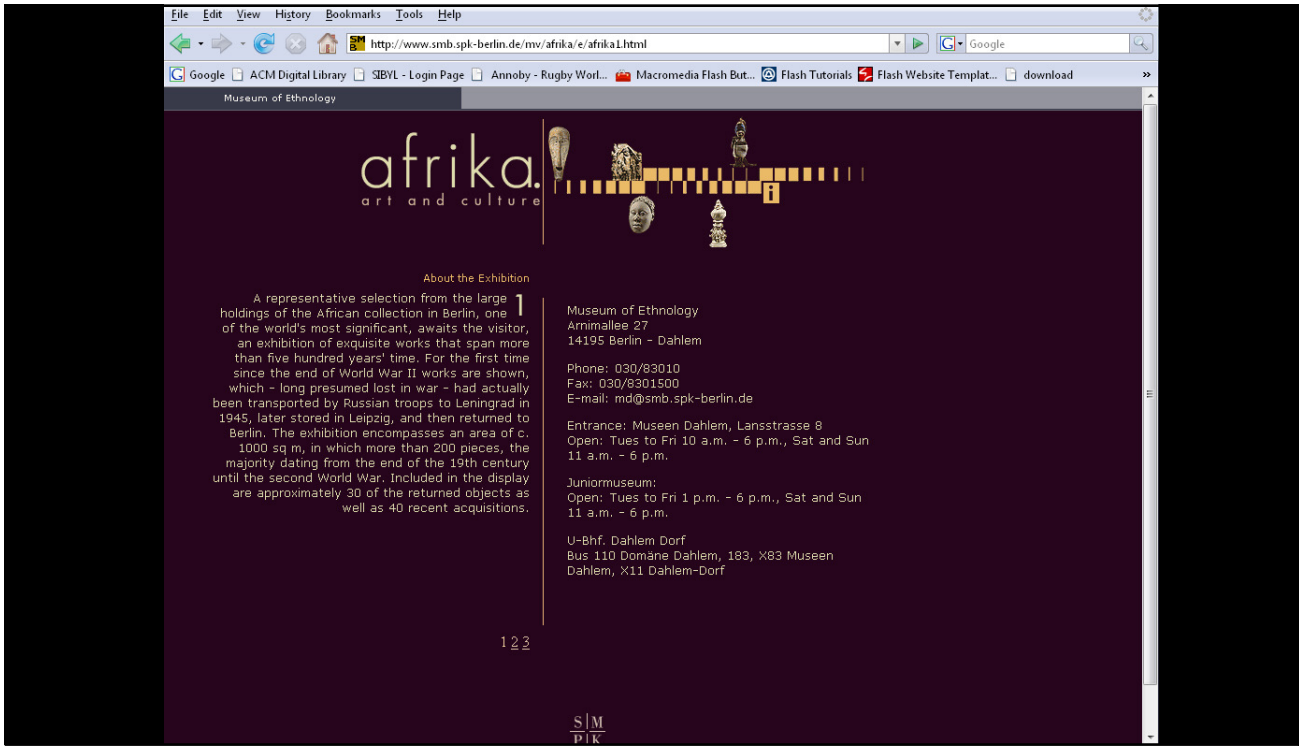
Designers can simply follow what empirical study results show

Be careful on the contrast difference between foreground and background…

White text
Yellow text
Red text
Blue text
Purple text
Green text

White text
Yellow text
Red text
Blue text
Purple text
Green text

23



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General Guidelines

Common sense, research findings, designer's experience, other disciplines… incorporating ***design principles***

Graphic Design

Use quiet background, so that what you want to **emphasise** can be emphasised

Sharpen the information

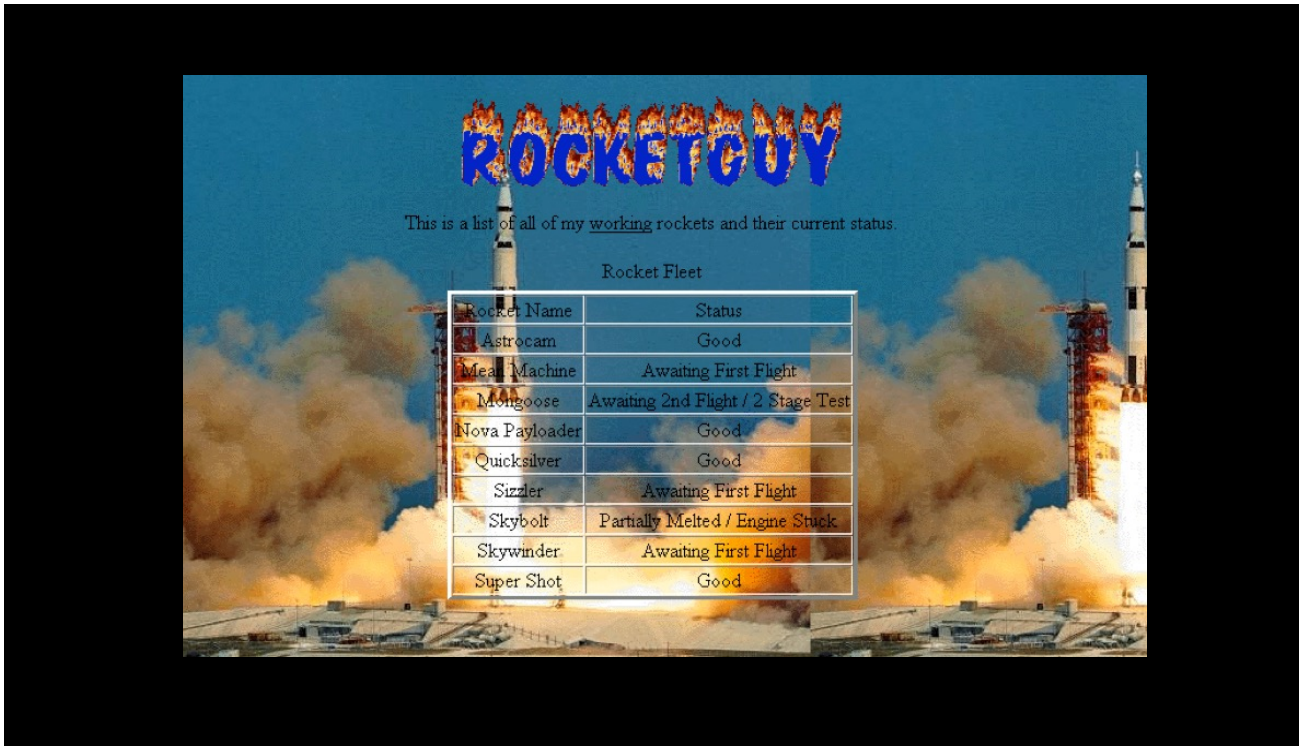
⋮

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General Guidelines

Common sense, research findings, designer's experience, other disciplines... incorporating ***design principles***

Graphic Design

Use quiet background, so that what you want to **emphasise** can be emphasised

Sharpen the information

⋮

➡ Applying design guidelines/principles will make a huge difference on your UI !

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General Guidelines

- Embedded design principles
- 40%+ of UI design textbooks
- 'Usability Principles', 'Heuristics', 'Design Rules', 'Design Guidelines'…

➡ We will deal with these for next 3 weeks

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WHAT WILL WE LEARN TODAY?

- Design Guidelines

… and associated concepts, principles and examples

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Housekeeping

Attend the classes!

- Important points in lecture
- Activities in class
- Lecture slides are not self-explanatory
- Cards will be distributed

➡ If you missed a class, check the lecture slides, ask colleagues who attended

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Human VS. Computer

Humans do better	Computers do better
Improvise	Calculate accurately
Relationship	Deductive reasoning A->B, B->C, A->C
Relevance and big picture, relating	Memory/storage
Inductive reasoning	Repetitive tasks
Creativity	Vigilance
Coping with unexpected situation	Endurance
:	:

Design knowledge (including principles & guidelines) incorporate these – check as/when we learn new design principles

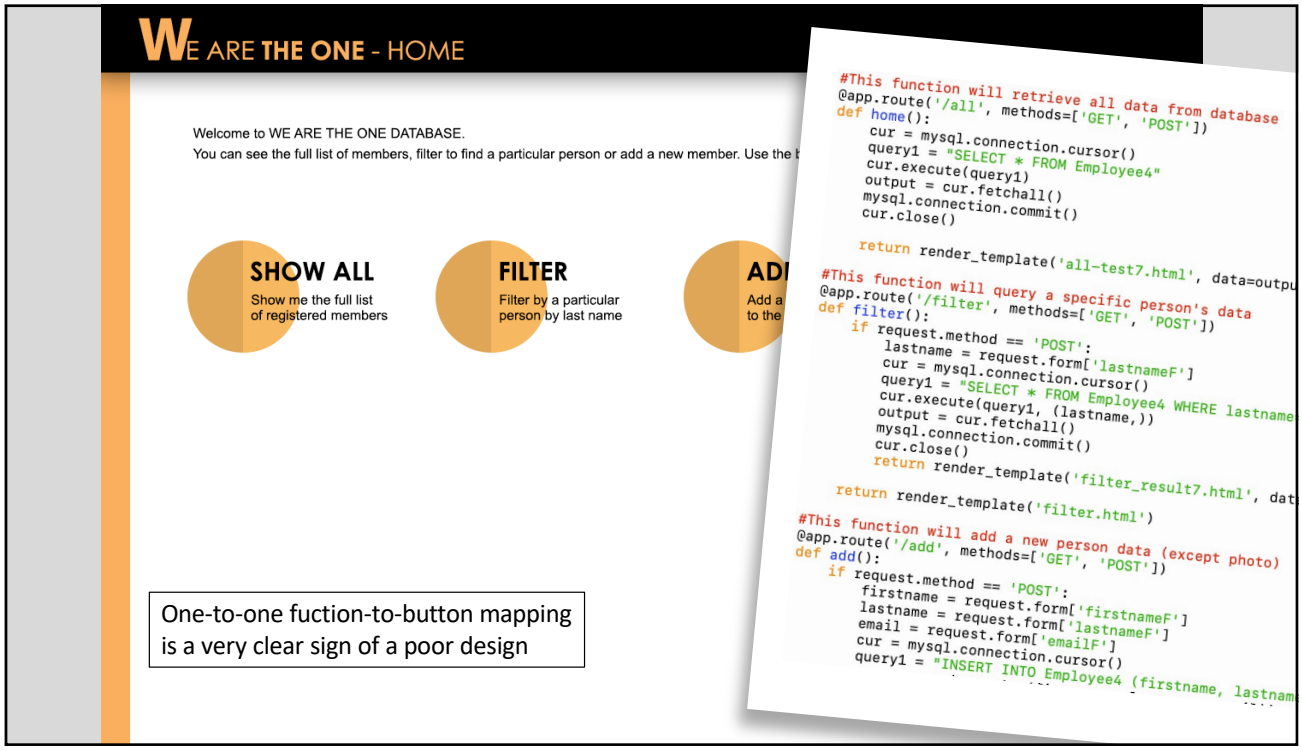
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Guideline 2: Simple and natural dialogue

- Minimise the *artefacts*
- Grouping – Separating - Prioritising - Sharpening

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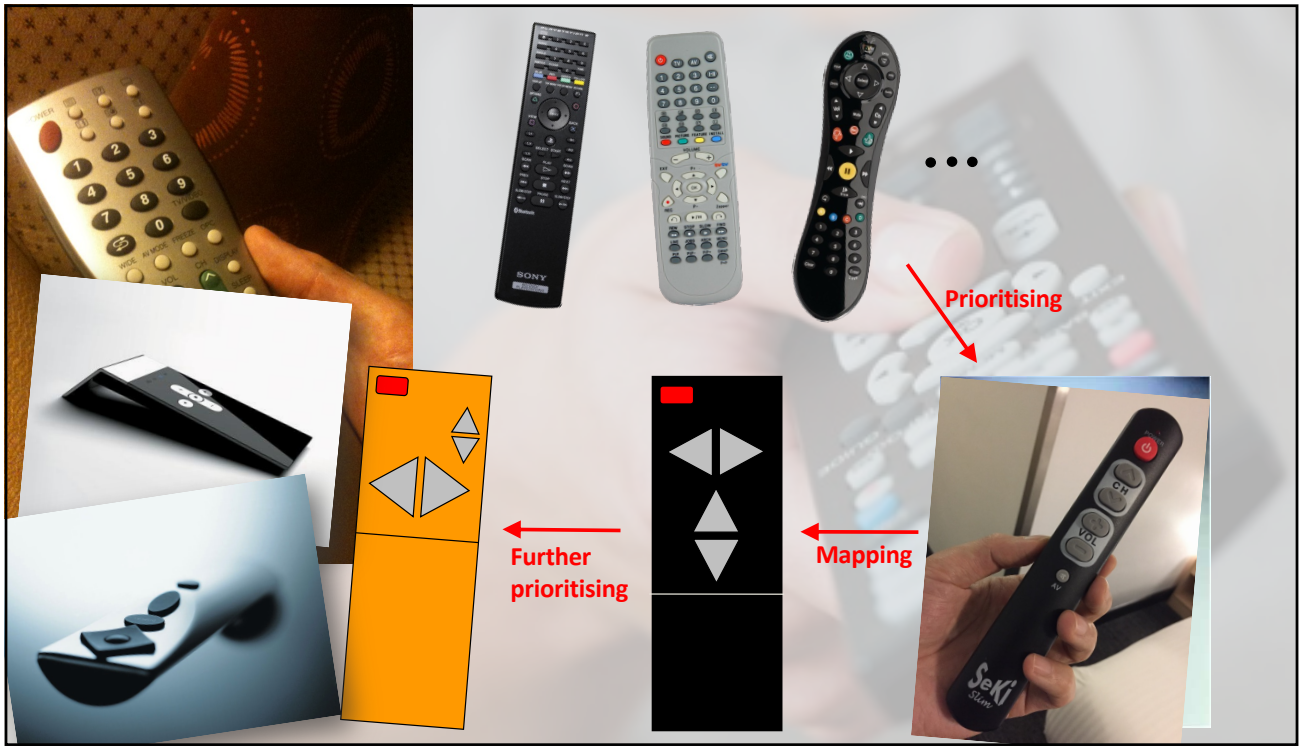
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WE ARE THE ONE - HOME
Welcome to WE ARE THE ONE DATABASE.
You can see the full list of members, filter to find a particular person or add a new member. Use the
SHOW ALL
Show me the full list of registered members
FILTER
Filter by a particular person by last name
ADD
Add a
to the
One-to-one fuction-to-button mapping is a very clear sign of a poor design
#This function will retrieve all data from database
@app.route('/all', methods=['GET', 'POST'])
def home():
cur = mysql.connection.cursor()
query1 = "SELECT * FROM Employee4"
cur.execute(query1)
output = cur.fetchall()
mysql.connection.commit()
cur.close()
return render_template('all-test7.html', data=outpu
#This function will query a specific person's data
@app.route('/filter', methods=['GET', 'POST'])
def filter():
if request.method == 'POST':
lastname = request.form['lastnameF']
cur = mysql.connection.cursor()
query1 = "SELECT * FROM Employee4 WHERE lastname
cur.execute(query1, (lastname,))
output = cur.fetchall()
mysql.connection.commit()
cur.close()
return render_template('filter_result7.html', dat
return render_template('filter.html')
#This function will add a new person data (except photo)
@app.route('/add', methods=['GET', 'POST'])
def add():
if request.method == 'POST':
firstname = request.form['firstnameF']
lastname = request.form['lastnameF']
email = request.form['emailF']
cur = mysql.connection.cursor()
query1 = "INSERT INTO Employee4 (firstname, lastnam

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WE ARE THE ONE - HOME
Welcome to WE ARE THE ONE DATABASE.
You can see the full list of members, filter to find a particular person or add a new member. Use the buttons below.
SHOW ALL
Show me the full list of registered members
FILTER
Filter by a particular person by last name
ADD NEW
Add a new member to the database
WE ARE THE ONE - HOME
428 members since Jan 2021
We have a fast-growing member base since January!
Browse and search our members and if you are new, register yourself in our social circle today.
Search a member
REGISTER
NEW MEMBERS
Janet
joined yesterday
Christopher
joined last week
Saibhan
joined last week
Think about why this is considered a better UI

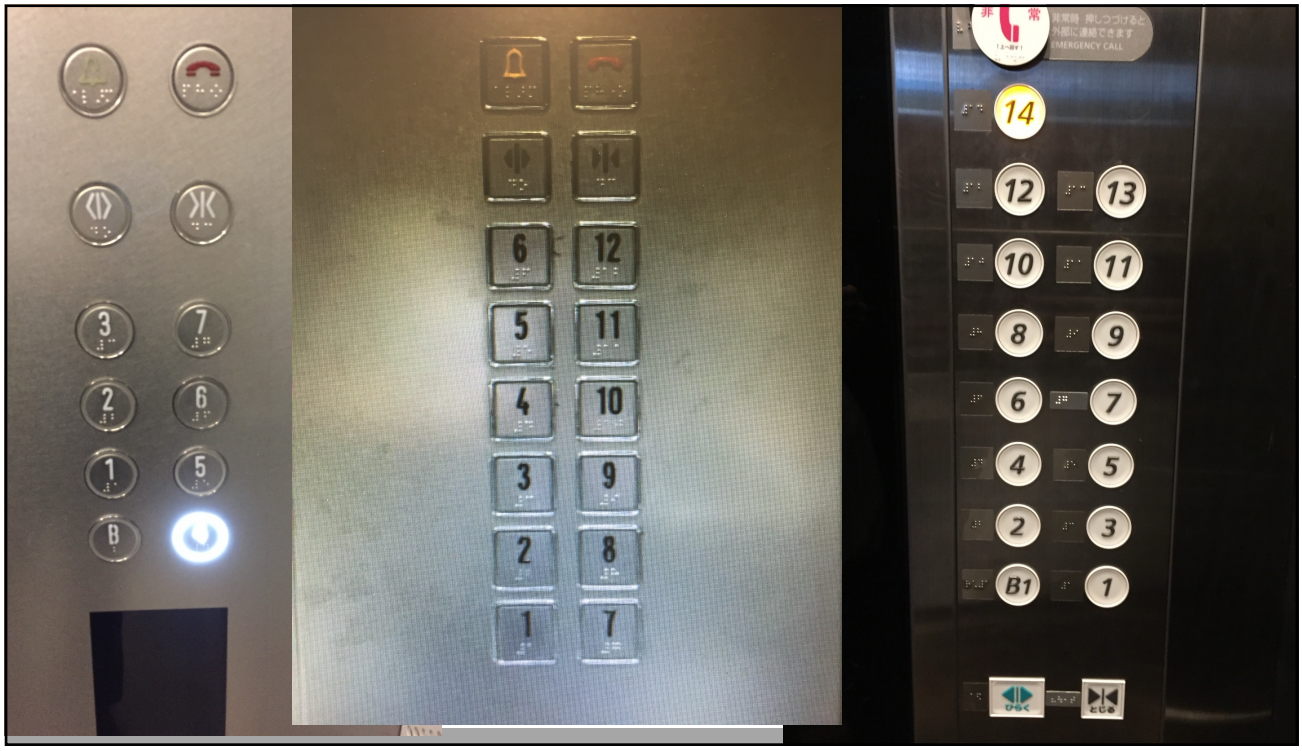
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Prioritising
Mapping
Further prioritising

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11
4
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7
G
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2
3
P4
P3
P2
P1
EMERGENCY TELEPHONE
PRESS BUTTON
FOR 5 SECONDS
AND AWAIT REPLY
L
LL
B2

38



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DESIGN EXERCISE 1

Re-design the lift control UI, based on

Grouping – Separating – Prioritising - Sharpening

16 levels + 4 underground levels



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Prototyping Tools

- Figma
- Marvel
- Origami
- FramerJS
- PowerPoint
- Photoshop
- :

List of prototyping tools and explanations:
https://slickplan.com/blog/best-prototyping-tools

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